

PROJECT REPORT ON
INVENTORY MANAGEMENT SYSTEM USING PYTHON,
TKINTER AND MYSQL

(Submitted in partial fulfilment of the requirement for the Internship Course

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By

DEV KUMAR AGGARWAL



ZillionSoftech Pvt.Ltd , Rewari

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CERTIFICATE

Certified that this Internship Report entitled “Inventory Management System using Python, Tkinter and MySQL” is an original report of the work done by me under the guidance of Project Mentor Mr Kamal Kumar, and under the supervision of Project Supervisor Ms Savita , Department of Computer Science, submitted as a part of the Project Course of the undergraduate Programme (B.tech Computer Science) of ZillionSoftech, Rewari

Dated: _____

Signature of Student: _____

Name: Dev Kumar Aggarwal

Countersigned by:

Project Supervisor

Department of Computer Science, ZillionSoftech,Rewari

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CHAPTER 1

INTRODUCTION

1.1 Background

Inventory management is one of the most important functions in any business organization. It involves managing product information, monitoring stock levels, maintaining supplier records, tracking purchases and sales, and generating financial reports. An efficient inventory management system helps organizations maintain optimal stock levels, reduce operational costs, and improve customer satisfaction by ensuring that products are available whenever required.

Traditionally, many small and medium-sized businesses manage inventory manually using paper registers or spreadsheet applications. Although these methods are inexpensive and easy to implement, they become inefficient as business operations expand. Manual inventory systems often result in duplicate records, inaccurate stock calculations, delayed report generation, and difficulty in tracking inventory movements.

With the rapid advancement of information technology, organizations have shifted toward computerized inventory management systems that automate routine business operations. Such systems improve data accuracy, reduce manual effort, provide real-time information, and generate analytical reports that support better business decisions.

The Enterprise Inventory & Financial Management System (Advanced) is developed to address these challenges. The application integrates inventory management with financial analysis, allowing administrators to manage products, suppliers, purchases, sales, expenses, and profitability from a single platform. Developed using Python, Tkinter, MySQL, and the Model–View–Controller (MVC) Architecture, the system provides a reliable, secure, and user-friendly solution for inventory management.

1.2 Existing System

Many organizations continue to use manual methods or spreadsheet-based applications for maintaining inventory records. While these methods are simple to use, they suffer from several limitations when handling large volumes of data.

The existing system has the following drawbacks:

Manual data entry increases the possibility of human errors.

Stock calculations consume considerable time.

Product information is difficult to search and maintain.

Financial reports must be prepared manually.

Low-stock products are not identified automatically.

Purchase and sales records become difficult to maintain accurately.

There is no centralized database for secure data storage.

Unauthorized users may access important records.

Business owners cannot monitor inventory performance in real time.

Report generation requires significant manual effort.

These limitations reduce business efficiency and increase operational costs.

1.3 Proposed System

The proposed system is a desktop-based Enterprise Inventory & Financial Management System that automates inventory operations and financial record management using modern software technologies.

The application provides secure administrator authentication and stores all business information within a centralized MySQL database. Users can efficiently add, update, delete, and search product records. Supplier information is also maintained digitally, allowing organizations to manage supplier directories effectively.

The system automatically records purchase and sales transactions, calculates revenue and expenses, and generates annual Profit & Loss reports. A graphical dashboard displays important business statistics such as total products, suppliers, revenue, expenses, purchase requests, and annual net profit.

Additionally, the application continuously monitors stock availability and generates low-stock alerts whenever product quantities fall below the predefined minimum level. This feature helps businesses avoid stock shortages and improves inventory planning.

Overall, the proposed system offers greater efficiency, improved accuracy, enhanced security, and better decision-making compared to traditional inventory management methods.

1.4 Problem Statement

Manual inventory management presents several operational challenges for businesses. Maintaining product information through paper records or spreadsheets often leads to duplicate entries, inaccurate stock calculations, delayed reporting, and inefficient inventory control.

Financial calculations require significant manual effort and are highly susceptible to computational errors. Tracking supplier information, monitoring stock availability, generating reports, and identifying products requiring immediate replenishment become increasingly difficult as business operations grow.

Therefore, there is a need for a computerized inventory and financial management system capable of automating inventory operations, improving data accuracy, reducing manual workload, and generating comprehensive business reports.

1.5 Scope of the Project

The scope of the Enterprise Inventory & Financial Management System includes complete management of inventory and financial operations for small and medium-sized businesses.

The application provides facilities for:

Secure administrator authentication

Product management

Product image management

Supplier management

Purchase record management

Sales transaction management

Dashboard analytics

Automatic revenue calculation

Expense management

Profit and loss analysis

Low stock monitoring

Purchase request generation

Password management

Theme customization

Secure logout

The system is designed primarily for desktop environments and is suitable for educational projects, retail stores, warehouses, and small-to-medium business organizations. It demonstrates the practical implementation of programming.

CHAPTER 2

OBJECTIVES

2.1 Primary Objective

The primary objective of the **Enterprise Inventory & Financial Management System (Advanced)** is to develop a secure, efficient, and user-friendly desktop application capable of managing inventory and financial activities within an organization. The system aims to automate inventory operations, reduce manual effort, improve data accuracy, and provide real-time business information to support effective decision-making. By integrating inventory management with financial reporting, the application offers a complete solution for managing products, suppliers, purchases, sales, expenses, and profitability.

2.2 Specific Objectives

The major objectives of this project are as follows:

1. Secure User Authentication

To develop a secure login system that allows only authorized users to access the application and protects sensitive business information from unauthorized access.

2. Product Management

To maintain complete product information including product name, category, brand, selling price, quantity, stock value, and product images through efficient Create, Read, Update, and Delete (CRUD) operations.

3. Supplier Management

To provide a centralized platform for storing and managing supplier information, making it easier to maintain supplier records and communicate with suppliers whenever required.

4. Purchase Management

To automate the recording of product purchases from suppliers, update inventory quantities automatically, and maintain complete purchase transaction records.

5. Sales Management

To record product sales efficiently, reduce available stock automatically after each sale, and maintain a detailed sales history for future reference.

6. Inventory Control

To maintain accurate inventory quantities and monitor product availability continuously, ensuring that stock levels remain updated after every purchase or sale transaction.

7. Revenue Calculation

To calculate annual revenue automatically based on sales transactions, eliminating manual calculations and improving financial accuracy.

8. Expense Management

To maintain complete records of purchase expenses and automatically calculate total business expenditures throughout the financial year.

9. Profit and Loss Analysis

To generate annual Profit & Loss reports by comparing total revenue with total expenses, enabling business owners to evaluate overall financial performance.

10. Dashboard Analytics

To provide a graphical dashboard displaying important business statistics such as total products, suppliers, purchase requests, revenue, expenses, and annual net profit for quick decision-making.

11. Low Stock Monitoring

To continuously monitor inventory levels and automatically generate low-stock alerts whenever product quantities fall below the predefined minimum level.

12. Purchase Request Generation

To generate purchase requests automatically for products that require replenishment, helping businesses avoid stock shortages.

13. User-Friendly Interface

To design an attractive and easy-to-use graphical user interface using Python Tkinter, enabling users with limited technical knowledge to operate the system efficiently.

14. Data Accuracy

To reduce manual errors through automated calculations, centralized data storage, and real-time database updates.

15. Database Management

To store all inventory and financial information securely in a MySQL relational database, ensuring consistency, integrity, and fast retrieval of records.

16. MVC Architecture Implementation

To implement the project using the Model–View–Controller (MVC) software architecture, improving code organization, maintainability, scalability, and modularity.

17. Practical Learning

To demonstrate practical knowledge of Python programming, Tkinter GUI development, MySQL database management, Object-Oriented Programming (OOP), and software engineering principles through the development of a real-world application.

2.3 Expected Outcomes

After successful implementation of the Enterprise Inventory & Financial Management System, the following outcomes are expected:

Improved inventory accuracy through automated stock management.

Reduced manual paperwork and operational workload.

Faster product, supplier, purchase, and sales management.

Secure storage of business information using MySQL.

Automatic generation of financial reports.

Improved decision-making through dashboard analytics.

Timely identification of low-stock products.

Better financial planning through accurate profit and loss analysis.

Enhanced user experience with a simple and interactive graphical interface.

A reliable, scalable, and maintainable desktop application suitable for educational purposes and small to medium-sized businesses.

CHAPTER 3

SOFTWARE REQUIREMENTS

3.1 Introduction

Software requirements define the applications, programming languages, libraries, and development tools required for designing, developing, testing, and executing the Enterprise Inventory & Financial Management System. Selecting appropriate software ensures that the application performs efficiently, remains secure, and is easy to maintain. This project has been developed using modern technologies that provide flexibility, reliability, and excellent support for desktop application development.

3.2 Software Requirements

The following software components were used during the development of the project.

Software component	Version
Operating System	Windows 11
Programming Language	Python 3.14
GUI Framework	TKinter
Database	MySQL 8.0
IDE	Visual Studio Code
Database Tool	MySQL Workbench
Image Processing Library	Pillow(PIL)
Version Control (Optional)	Git

3.3 Development Tools

Windows 11

Windows 11 serves as the operating system used for developing and testing the application. It provides a stable and secure environment for Python programming and supports all the software required for this project.

Python 3.14

Python is the primary programming language used in the project. It is simple, powerful, and supports object-oriented programming. Python provides extensive libraries that

simplify application development, database connectivity, and graphical user interface design.

Tkinter

Tkinter is Python's standard Graphical User Interface (GUI) library. It is used to design windows, forms, buttons, labels, menus, tables, and dialog boxes. The graphical interface developed using Tkinter makes the application easy to operate and improves user experience.

MySQL 8.0

MySQL is an open-source Relational Database Management System (RDBMS) used for storing and managing all project data. It maintains product details, supplier records, purchase history, sales transactions, login credentials, and financial information securely.

Visual Studio Code

Visual Studio Code is the Integrated Development Environment (IDE) used to write, edit, debug, and manage the project's source code. It offers syntax highlighting, intelligent code completion, debugging tools, and Git integration, making software development more efficient.

MySQL Workbench

MySQL Workbench is used to create, manage, and maintain the MySQL database. It provides a graphical interface for designing database tables, executing SQL queries, and monitoring database performance.

Pillow (PIL)

Pillow is an image processing library used for handling product images within the application. It allows the system to load, resize, and display images, improving the visual appearance and usability of the Product Catalog module.

Git

Git is an optional version control system that helps developers maintain different versions of the project. It tracks code changes, supports collaboration, and allows recovery of previous versions when required.

3.4 Importance of Software Requirements

The selected software technologies provide several advantages during the development and execution of the project:

Easy application development using Python.

Interactive graphical interface using Tkinter.

Secure and efficient database management using MySQL.

Simplified coding and debugging with Visual Studio Code.

Reliable database administration using MySQL Workbench.

Efficient image management using Pillow.

Better project management through Git version control.

High compatibility with Windows operating systems.

Improved scalability and maintainability of the application.

3.5 Summary

The Enterprise Inventory & Financial Management System has been developed using reliable and widely accepted software technologies. Python, Tkinter, and MySQL form the core development platform, while Visual Studio Code and MySQL Workbench simplify programming and database management. These tools collectively provide a stable, secure, and efficient environment for developing a professional desktop-based inventory and financial management application.

CHAPTER 4

HARDWARE REQUIREMENTS

4.1 Introduction

Hardware requirements specify the minimum and recommended computer specifications needed to develop and run the Enterprise Inventory & Financial Management System efficiently. Since the application is a desktop-based software developed using Python, Tkinter, and MySQL, it does not require high-end hardware. A standard personal computer with moderate specifications is sufficient for development, testing, and deployment.

Proper hardware ensures smooth execution of the application, faster database operations, better graphical performance, and an improved user experience.

4.2 Minimum Hardware Requirements

The following hardware configuration is sufficient for running the application.

Processor	Intel core i3 or Higher
RAM	4 GB Minimum
Storage	500 MB Free Space
Display	1366 × 768 Resolution
Keyboard	Standard Keyboard
Mouse	Optical Mouse
Internet	Required only for software installation

4.3 Recommended Hardware Configuration

For better performance and faster processing, the following hardware configuration is recommended.

Component	Recommended Specification
Processor	Intel Core i5 / AMD Ryzen 5 or Higher
RAM	8 GB or More
Storage	SSD (Solid State Drive)
Operating System	Windows 11
Display	Full HD (1920 × 1080)
Graphics	Integrated or Dedicated Graphics (Optional)
Internet	Broadband Connection for Updates

4.4 Hardware Description

Processor

The processor executes all instructions of the application. A modern multi-core processor improves application speed, database processing, and multitasking performance.

RAM (Random Access Memory)

RAM stores temporary data while the application is running. A minimum of 4 GB RAM allows the application to operate smoothly, while 8 GB RAM provides better multitasking and improved responsiveness.

Storage Device

The application requires only a small amount of storage space. However, using an SSD significantly reduces application startup time, improves database performance, and increases overall system responsiveness.

Display

A display resolution of at least **1366 × 768** is required to properly view the graphical user interface. A Full HD display provides better clarity and allows more information to be displayed simultaneously.

Keyboard and Mouse

A standard keyboard is used for entering product information, supplier details, and financial records, while an optical mouse provides convenient navigation through the graphical user interface.

Internet Connection

An internet connection is not necessary for the daily operation of the application. It is required only for downloading development tools, software updates, or future cloud-based enhancements.

4.5 Importance of Hardware Requirements

The selected hardware configuration provides several benefits:

Smooth execution of the desktop application.

Faster loading of product and supplier records.

Improved database processing speed.

Better graphical interface performance.

Reduced application response time.

Efficient multitasking during development and testing.

Reliable long-term performance.

Improved user experience.

4.6 Summary

The Enterprise Inventory & Financial Management System is designed to run efficiently on standard desktop and laptop computers. Its modest hardware requirements make it suitable for educational institutions, retail stores, warehouses, and small to medium-sized businesses. The recommended configuration provides enhanced speed, stability, and reliability while ensuring that the application performs smoothly during inventory and financial management operations.

CHAPTER 5

TECHNOLOGIES USED

5.1 Introduction

The Enterprise Inventory & Financial Management System has been developed using modern software technologies that ensure reliability, security, efficiency, and ease of maintenance. Each technology used in this project plays a significant role in developing a user-friendly desktop application capable of managing inventory, suppliers, purchases, sales, and financial records. The combination of Python, Tkinter, MySQL, MVC Architecture, Object-Oriented Programming, and Pillow provides a strong foundation for building a scalable and maintainable system.

5.2 Python

Python is a high-level, interpreted, and object-oriented programming language widely used for desktop applications, web development, artificial intelligence, automation, and data analysis. It is known for its simple syntax, readability, and extensive standard libraries.

In this project, Python serves as the core programming language. It is responsible for implementing the business logic, processing user requests, performing calculations, validating data, interacting with the database, and controlling the overall workflow of the application.

Advantages of Python

Simple and easy-to-understand syntax.

Platform-independent programming language.

Supports object-oriented programming.

Large collection of built-in libraries.

Faster development process.

Excellent database connectivity.

Highly maintainable and scalable.

5.3 Tkinter

Tkinter is Python's standard Graphical User Interface (GUI) library used for designing desktop applications. It provides various graphical components such as windows, buttons, labels, menus, frames, text boxes, tables, and dialog boxes.

The Enterprise Inventory & Financial Management System uses Tkinter to create an attractive, interactive, and user-friendly interface. Every module of the application, including Login, Dashboard, Product Management, Supplier Management, Purchase Records, Sales History, Profit & Loss Reports, and Settings, is designed using Tkinter.

Advantages of Tkinter

Easy GUI development.

Built into Python.

Lightweight and fast.

Cross-platform compatibility.

Supports event-driven programming.

Easy integration with Python code.

5.4 MySQL

MySQL is an open-source Relational Database Management System (RDBMS) used for storing and retrieving application data efficiently. It organizes information into tables and allows fast execution of SQL queries.

In this project, MySQL stores administrator credentials, product records, supplier details, purchase history, sales transactions, financial data, and system settings. It ensures data consistency, integrity, and secure storage.

Advantages of MySQL

High performance.

Secure data storage.

Multi-user support.

Data consistency.

Fast query execution.

Backup and recovery features.

Scalability.

Open-source availability.

5.5 Model–View–Controller (MVC) Architecture

The Enterprise Inventory & Financial Management System follows the Model–View–Controller (MVC) Architecture, a software design pattern that separates an application into three independent components.

Model

The Model manages all database operations and communicates directly with the MySQL database. It performs Create, Read, Update, and Delete (CRUD) operations while maintaining data integrity.

View

The View represents the graphical user interface developed using Tkinter. It displays forms, tables, buttons, menus, reports, and other visual components that allow users to interact with the application.

Controller

The Controller acts as the bridge between the View and the Model. It receives user input, validates the data, executes business logic, communicates with the database through the Model, and returns the processed results to the View.

Advantages of MVC

Better code organization.

Easy debugging.

Improved scalability.

High code reusability.

Easier maintenance.

Supports modular programming.

Simplifies future enhancements.

MVC STRUCTURE

```
InventoryManagementSystem/
|
├── main.py
├── config.py
├── requirements.txt
├── README.md
|
├── assets/
|
├── controllers/
|   ├── __init__.py
|   ├── dashboard_controller.py
|   ├── fee_controller.py
|   ├── inventory_controller.py
|   ├── login_controller.py
|   ├── product_controller.py
|   ├── purchase_controller.py
|   ├── sale_controller.py
|   └── supplier_controller.py
|
├── dao/
|   ├── __init__.py
|   ├── product_dao.py
|   ├── purchase_dao.py
|   ├── sale_dao.py
|   ├── supplier_dao.py
|   └── user_dao.py
|
├── database/
|   ├── __init__.py
|   ├── backup.py
|   ├── create_database.py
|   ├── db_connection.py
|   └── sql/
|       └── inventory_db.sql
|
├── models/
|   ├── __init__.py
|   ├── product.py
|   ├── purchase.py
|   ├── sale.py
|   ├── supplier.py
|   └── user.py
```

```
├── services/
│   ├── __init__.py
│   ├── auth_service.py
│   ├── export_service.py
│   ├── report_service.py
│   └── validation_service.py
├── utils/
│   ├── __init__.py
│   ├── constants.py
│   ├── helper.py
│   ├── messagebox_util.py
│   └── validator.py
├── views/
│   ├── __init__.py
│   ├── dashboard.py
│   ├── login.py
│   ├── settings.py
│   ├── product/
│   │   ├── __init__.py
│   │   ├── product_catalog.py
│   │   └── product_image.py
│   ├── purchase/
│   │   ├── __init__.py
│   │   └── purchases.py
│   ├── sale/
│   │   ├── __init__.py
│   │   └── sales.py
│   ├── supplier/
│   │   ├── __init__.py
│   │   └── suppliers.py
│   └── report/
│       ├── __init__.py
│       ├── low_stock.py
│       └── profit_loss.py
├── reports/
└── logs/
```

```
└─ tests/  
    └─ __init__.py
```

5.6 Object-Oriented Programming (OOP)

The project has been developed using Object-Oriented Programming principles. OOP organizes the software into classes and objects, making the code more structured, reusable, and easier to maintain.

The application uses concepts such as:

Classes

Objects

Inheritance

Encapsulation

Polymorphism

Modular Programming

These concepts improve software quality and reduce code duplication while simplifying future development.

Advantages of OOP

Improved code reusability.

Better maintainability.

Simplified debugging.

Modular application design.

Easy scalability.

Reduced development time.

5.7 Pillow (PIL)

Pillow (Python Imaging Library) is an image-processing library used for handling product images within the application. It allows images to be loaded, resized, displayed, and managed efficiently.

In the Product Catalog module, Pillow enables administrators to upload product images that help identify inventory items visually. This feature improves usability and enhances the overall appearance of the application.

Advantages of Pillow

Efficient image loading.

Image resizing support.

High-quality image display.

Easy integration with Tkinter.

Improved graphical presentation.

Better user experience.

5.8 DAO (Data Access Object)

The project also implements the DAO design pattern to isolate all database operations from the rest of the application. Instead of writing SQL queries directly in the GUI or controller, all database communication is performed through dedicated DAO classes.

DAO Classes

ProductDAO

SupplierDAO

PurchaseDAO

SaleDAO

UserDAO

Each DAO class is responsible for performing CRUD operations related to its corresponding entity.

Advantages of DAO

Centralized database access.

Improved security.

Easier maintenance.

Better code organization.

Reusable SQL operations.

Simplified testing.

DAO Workflow

Controller receives a user request.

Controller calls the appropriate DAO method.

DAO executes SQL queries.

Results are returned to the Controller.

The View displays the updated information.

5.8 Summary

The technologies used in this project provide a reliable and efficient platform for developing a professional desktop application. Python provides the application logic, Tkinter creates the graphical interface, MySQL manages data storage, MVC Architecture organizes the software structure, Object-Oriented Programming improves maintainability, and Pillow enhances image management. Together, these technologies ensure that the Enterprise Inventory & Financial Management System is secure, scalable, user-friendly, and capable of meeting the inventory and financial management needs of small and medium-sized organizations.

CHAPTER 6

SYSTEM ARCHITECTURE

6.1 Introduction

System architecture defines the overall structure and organization of the Enterprise Inventory & Financial Management System. It describes how different software components interact with each other to perform various business operations. A well-designed architecture improves software quality by making the application easier to understand, develop, test, maintain, and upgrade.

This project follows the Model–View–Controller (MVC) architecture. The MVC design pattern separates the application into three independent components: Model, View, and Controller. This separation allows each component to perform a specific responsibility without affecting the others.

The user interacts with the graphical interface (View), which forwards user requests to the Controller. The Controller processes these requests, communicates with the Model to retrieve or update information stored in the MySQL database, and finally sends the processed results back to the View for display.

6.2 System Architecture Overview

The Enterprise Inventory & Financial Management System consists of the following major components:

User Interface (Tkinter)

Controller Layer

Business Logic

Database Layer (MySQL)

Image Management (Pillow)

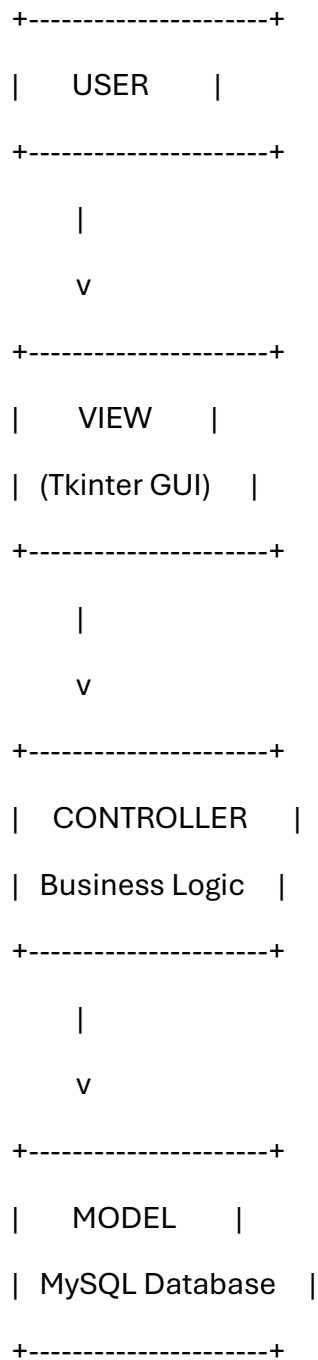
Reporting Module

Dashboard Analytics

Each component performs a dedicated function while communicating efficiently with other modules to ensure smooth system operation.

6.3 MVC Architecture Diagram

Figure 6.1: MVC Architecture of the Enterprise Inventory & Financial Management System



6.4 Components of MVC Architecture

Model

The Model is responsible for managing all database-related operations. It directly communicates with the MySQL database to store, retrieve, update, and delete records. All SQL queries and database connections are handled within the Model layer.

The responsibilities of the Model include:

Product management

Supplier management

Sales records

Purchase records

User authentication

Financial calculations

Database connectivity

The Model ensures that all application data remains accurate, secure, and consistent.

View

The View represents the graphical user interface (GUI) developed using Python Tkinter. It provides interactive windows, forms, tables, buttons, menus, and reports that allow users to communicate with the system.

The View contains:

Login Screen

Dashboard

Product Catalog

Supplier Directory

Purchase Module

Sales History

Profit & Loss Report

Settings Module

The View only displays information and collects user input. It does not directly access the database.

Controller

The Controller acts as the communication bridge between the View and the Model. It receives requests from the user interface, validates user input, executes business logic, and communicates with the Model to perform database operations.

Major responsibilities include:

Login verification

Product validation

Supplier management

Purchase processing

Sales processing

Revenue calculation

Expense calculation

Dashboard updates

Low stock monitoring

Report generation

The Controller improves code organization by separating business logic from the graphical interface.

6.5 Workflow of the System

The overall workflow of the application is as follows:

The administrator opens the application.

The Login module verifies user credentials.

After successful authentication, the Dashboard is displayed.

The administrator manages products and suppliers.

Purchase transactions update inventory quantities.

Sales transactions reduce available stock.

Revenue and expense calculations are performed automatically.

The Dashboard updates business statistics in real time.

The Low Stock Alert module generates notifications whenever inventory falls below the minimum level.

The Profit & Loss module prepares annual financial reports.

This workflow ensures that inventory operations and financial records remain synchronized throughout the application.

6.6 Advantages of MVC Architecture

The MVC architecture provides several important advantages:

Better code organization.

Easy debugging and testing.

High code reusability.

Improved maintainability.

Modular software development.

Easy implementation of new features.

Reduced code duplication.

Better collaboration among developers.

Improved software scalability.

Faster application development.

6.7 Summary

The Enterprise Inventory & Financial Management System follows the Model–View–Controller (MVC) architecture to provide a structured and maintainable software design. The separation of responsibilities among the Model, View, and Controller improves software quality, simplifies maintenance, and supports future enhancements. The architecture ensures efficient communication between the graphical interface and the database while maintaining data integrity and providing a smooth user experience.

CHAPTER 7

DATABASE DESIGN

7.1 Introduction

A database is the backbone of any inventory management system. It stores all business information in an organized and structured manner, allowing users to retrieve, update, and manage records efficiently. The Enterprise Inventory & Financial Management System uses MySQL, a relational database management system (RDBMS), to maintain product information, supplier details, purchase records, sales transactions, user credentials, and financial reports.

The database has been designed to reduce redundancy, improve data consistency, and maintain the integrity of business information. Each table represents a specific entity, and relationships between tables enable efficient data management.

7.2 Objectives of Database Design

The main objectives of the database design are:

- To store inventory data securely.
 - To maintain product and supplier information.
 - To record purchase and sales transactions.
 - To support user authentication.
 - To generate financial reports.
 - To reduce data redundancy.
 - To ensure data consistency and integrity.
 - To improve query performance and data retrieval.
 - To support future expansion of the application.
-

7.3 Database Tables

The system consists of major database tables.

1 User Table
2 Product
3 Supplier
4 Purchase
5 Sales
6 Primary Keys

1. Users Table

Purpose:

Stores administrator login credentials and user information.

Field Name	Data Type	Description
id	INT	Primary Key
Username	VARCHAR	Login Username
Password	VARCHAR	Encrypted Password
Role	VARCHAR	User Role

2. Product Table

Purpose:

Stores complete details of all products available in inventory.

Field Name	Data Type	Description
Product_id	INT	Primary Key
Product_name	VARCHAR	Product Name

Brand	VARCHAR	Brand Name
Category	VARCHAR	Product Category
Selling_Price	DECIMAL	Selling Price
Quantity	INT	Available Stock
Total_value	DECIMAL	Total Stock Value
Image_path	VARCHAR	Product Image Path

3. Supplier Table

Purpose:

Stores supplier information.

Field Name	Data Type	Description
Supplier_id	INT	Primary Key
Supplier_name	VARCHAR	Company Name
Contact_number	VARCHAR	Mobile Address
Email	VARCHAR	Email Address
Field Name	Data Type	Description
Address	TEXT	Supplier Address

4. Purchase Table

Purpose:

Maintains all inventory purchase transactions.

Field Name	Data Type	Description
Purchase_id	INT	Primary Key
Product_name	VARCHAR	Purchased Product
Supplier_name	VARCHAR	Supplier Name
Cost_Price	VARCHAR	Cost Price
Quantity	INT	Purchased Quantity
Purchase_Date	DATE Time	Purchase date Time

5. Sales Table

Purpose:

Stores all sales transactions.

Field Name	Data Type	Description
Sale_id	INT	Primary Key
Product_name	VARCHAR	Sold Product
Selling_price	DECIMAL	Selling Price
Quantity_sold	INT	Quantity sold
Total_revenue	DECIMAL	Revenue Generated
Sale_date	DATE TIME	Sale Date Time

7.4 Database Relationship

The relationship among different database tables is illustrated below.

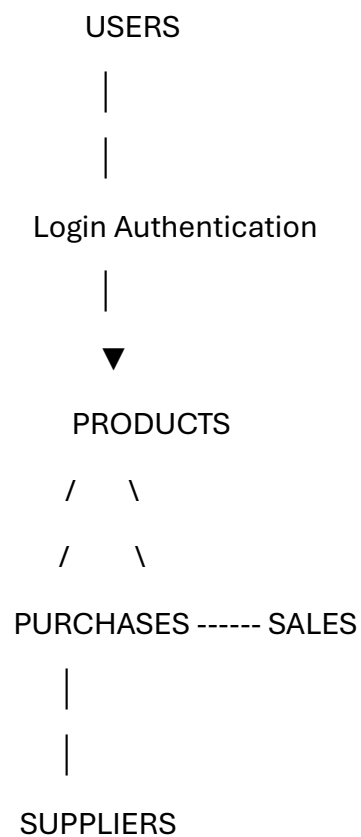


Figure 7.1: Relationship between database tables.

This structure ensures proper communication between inventory, supplier, purchase, and sales information while maintaining data consistency.

7.5 Normalization

To improve database efficiency, normalization principles have been applied during database design.

First Normal Form (1NF)

Each table contains atomic values.

Duplicate columns are eliminated.

Every record is uniquely identified.

Second Normal Form (2NF)

All non-key attributes depend on the primary key.

Partial dependency is removed.

Third Normal Form (3NF)

Transitive dependencies are eliminated.

Data redundancy is minimized.

Data integrity is improved.

Applying normalization reduces unnecessary duplication and improves overall database performance.

7.6 Primary Keys

Each table contains a unique primary key that uniquely identifies every record.

Table	Primary Key
Users	ID
Products	Product_id
Suppliers	Supplier_id
Purchases	Purchase_id
Sales	Sale_id

Primary keys prevent duplicate records and improve query performance.

7.7 Advantages of Using MySQL

The Enterprise Inventory & Financial Management System uses MySQL because it offers numerous advantages.

High performance and speed.

Secure storage of business information.

Multi-user support.

Fast SQL query execution.

Reliable backup and recovery.

Data consistency and integrity.

Open-source availability.

Easy integration with Python.

Scalability for future enhancements.

7.8 Database Security

The database incorporates several security measures to protect business information.

Administrator authentication before access.

Secure storage of login credentials.

Restricted access to sensitive data.

Input validation before database operations.

Controlled database connectivity.

Backup support for data recovery.

These security features help maintain confidentiality, integrity, and availability of business information.

7.9 Summary

The database design of the Enterprise Inventory & Financial Management System provides a structured and efficient method for storing and managing inventory and financial information. Using MySQL, normalized tables, primary keys, and secure

relationships ensures reliable data storage, quick retrieval, and smooth execution of all application modules. The database architecture forms the foundation of the entire system and supports accurate inventory management, financial reporting, and future system scalability.

CHAPTER 8

LOGIN MODULE

8.1 Introduction

The Login Module is the first and one of the most important components of the Enterprise Inventory & Financial Management System. It provides a secure authentication mechanism that allows only authorized users to access the application. Before entering the system, users must provide a valid username and password. If the credentials match the records stored in the MySQL database, access is granted; otherwise, an error message is displayed.

This module protects confidential business information from unauthorized access and serves as the security gateway for all other modules.

8.2 Objectives of the Login Module

The objectives of the Login Module are:

- To authenticate authorized users.
 - To prevent unauthorized system access.
 - To protect inventory and financial records.
 - To verify user credentials using the MySQL database.
 - To improve system security and reliability.
 - To provide a simple and user-friendly login interface.
-

8.3 Features of the Login Module

The Login Module provides the following features:

- Username and password authentication.
- Secure login verification.
- Password masking for privacy.
- Error messages for invalid credentials.
- Reset and clear input fields.
- Easy navigation to the dashboard after successful login.

Database-based user authentication.

Fast and responsive login process.

8.4 Working Process

The Login Module follows the steps below:

The user starts the application.

The Login window is displayed.

The user enters the username and password.

The Controller validates the entered information.

The Model checks the credentials stored in the MySQL database.

If the credentials are correct, the Dashboard window opens.

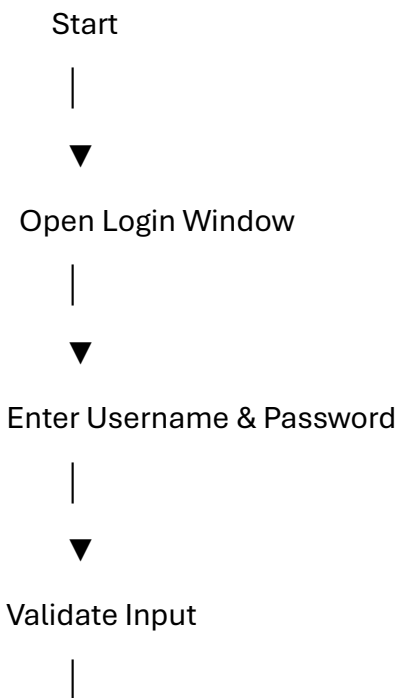
If the credentials are incorrect, an error message is displayed.

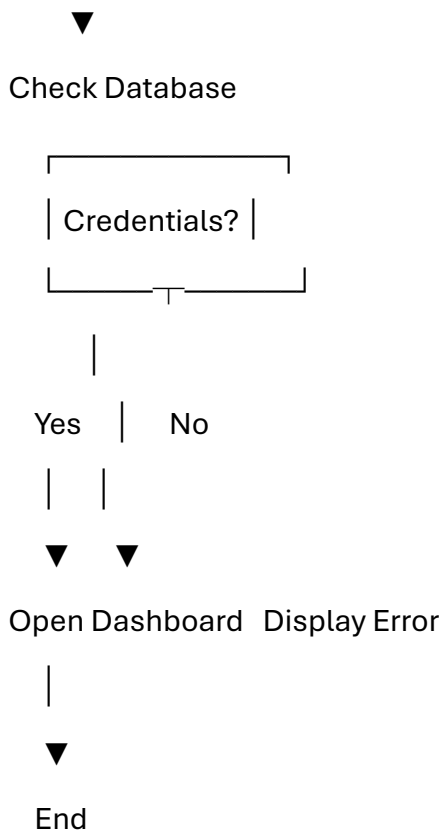
The user can retry with valid credentials.

This process ensures that only authorized users can access the application's features.

8.5 Login Flow Diagram

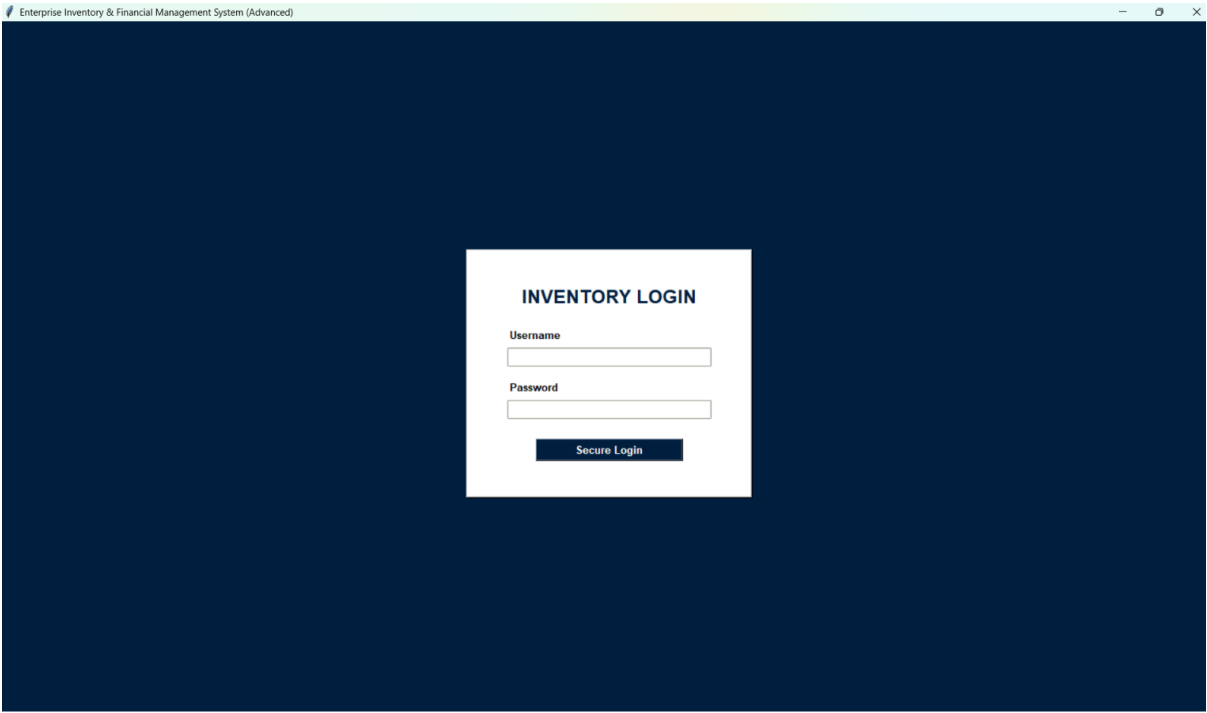
Figure 8.1: Login Module Flow





8.6 Login Screen

Figure 8.2: Login Screen



Screenshot Description:

The Login screen contains:

Application title.

Username field.

Password field.

Login button.

Clear button (if implemented).

Exit button.

Attractive graphical user interface developed using Tkinter.

The interface is simple, responsive, and easy to use, ensuring a smooth authentication process.

8.7 Advantages of the Login Module

The Login Module offers several advantages:

Prevents unauthorized access.

Protects sensitive inventory and financial information.

Provides secure authentication.

Easy to use.

Fast login process.

Improves overall system security.

Supports database-based verification.

Reduces the risk of unauthorized modifications.

8.8 Summary

The Login Module acts as the security gateway of the Enterprise Inventory & Financial Management System. By verifying user credentials through the MySQL database, it ensures that only authorized users can access the application. The module is designed with a user-friendly interface, quick authentication process, and strong security measures, making it a reliable foundation for the entire system.

CHAPTER 9

DASHBOARD MODULE

9.1 Introduction

The Dashboard Module is the central component of the Enterprise Inventory & Financial Management System. It provides administrators with a complete overview of inventory and financial information immediately after a successful login. Instead of opening multiple modules individually, users can monitor important business statistics from a single screen.

The dashboard displays real-time information such as the total number of products, suppliers, purchase requests, annual revenue, annual expenses, and annual net profit. It also provides quick navigation to different modules of the application, making business management more efficient.

9.2 Objectives of the Dashboard Module

The objectives of the Dashboard Module are:

- To provide a quick overview of business performance.
 - To display important inventory statistics.
 - To present financial information in real time.
 - To reduce the time required to access business data.
 - To improve decision-making through summarized reports.
 - To provide easy navigation to all system modules.
-

9.3 Features of the Dashboard

The Dashboard includes the following features:

- Total Products counter.
- Total Suppliers counter.
- Purchase Requests counter.
- Annual Revenue display.
- Annual Expenses display.

Annual Net Profit display.

Quick access navigation buttons.

Date and time display.

Attractive graphical interface.

Real-time data updates.

These features enable administrators to monitor business performance efficiently without opening multiple windows.

9.4 Working Process

The Dashboard Module works as follows:

The administrator logs into the system successfully.

The Dashboard window opens automatically.

The Controller retrieves information from the MySQL database.

The Dashboard displays current inventory statistics.

Revenue and expense values are calculated automatically.

Net Profit is displayed using the formula:

Net Profit = Total Revenue – Total Expenses

Purchase requests and low-stock products are updated dynamically.

The administrator can navigate to any module directly from the dashboard.

9.5 Dashboard Components

The dashboard consists of several information cards that display business statistics.

Total Products

Displays the total number of products available in the inventory.

Total Suppliers

Shows the total number of suppliers registered in the system.

Purchase Requests

Displays products that require replenishment due to low stock.

Annual Revenue

Shows the total revenue generated through product sales.

Annual Expenses

Displays the total amount spent on purchasing inventory.

Annual Net Profit

Calculates the organization's annual profit after deducting expenses from revenue.

Navigation Panel

Provides quick access to:

Product Catalog

Supplier Directory

Purchase Module

Sales History

Profit & Loss

Settings

Logout

9.6 Dashboard Screenshot

Figure 9.1: Dashboard Window

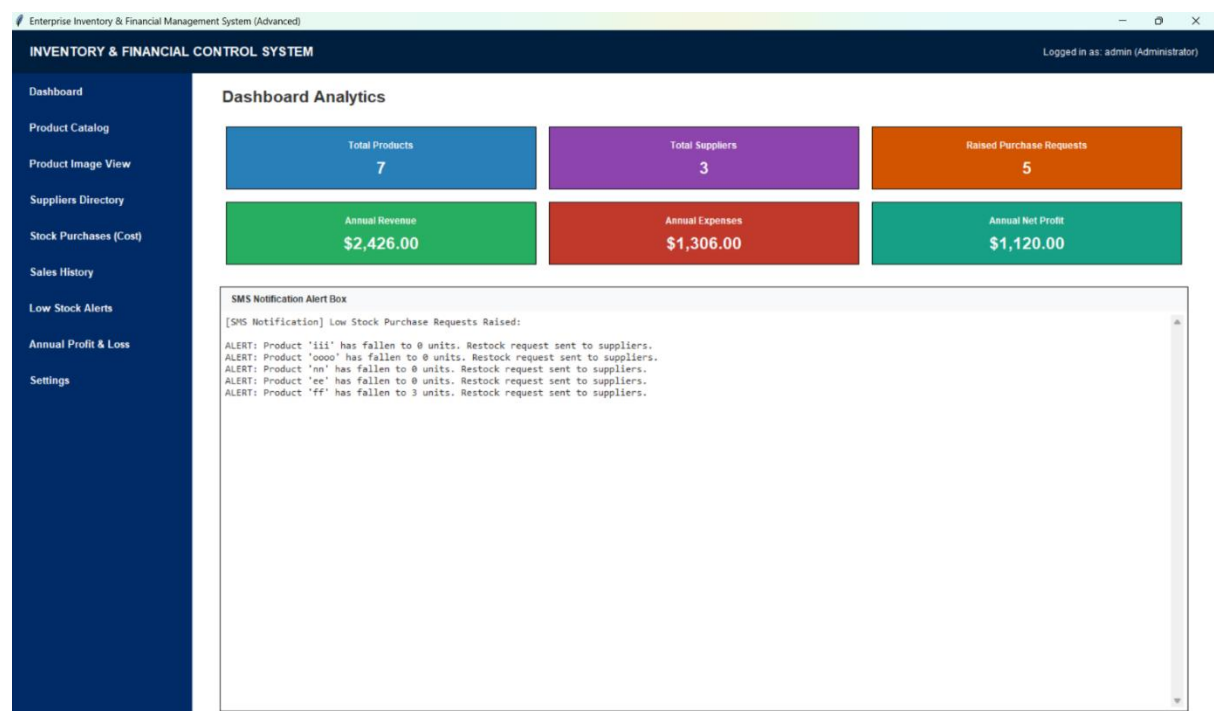


Figure Description

The Dashboard displays:

Total Products

Total Suppliers

Purchase Requests

Annual Revenue

Annual Expenses

Annual Net Profit

Navigation Menu

Current Date and Time

The interface is clean, responsive, and designed to help administrators monitor business activities efficiently.

9.7 Advantages of the Dashboard Module

The Dashboard Module provides several advantages:

Displays business information in one place.

Saves time by reducing navigation.

Supports real-time monitoring.

Improves decision-making.

Provides instant financial analysis.

Enhances productivity.

Offers a professional graphical interface.

Easy to understand and operate.

9.8 Summary

The Dashboard Module serves as the central monitoring system of the Enterprise Inventory & Financial Management System. By presenting inventory statistics, financial information, and navigation options on a single screen, it enables administrators to

monitor business performance effectively. The dashboard simplifies inventory management, supports faster decision-making, and improves overall operational efficiency.

CHAPTER 10

PRODUCT CATALOG MODULE

10.1 Introduction

The Product Catalog Module is one of the most important components of the Enterprise Inventory & Financial Management System. It enables administrators to maintain complete information about all products available in the inventory. This module provides facilities to add new products, update existing product details, delete unwanted records, search for products, and manage product images.

By maintaining accurate product information, the module helps improve inventory control, reduces manual errors, and ensures that stock records remain up to date. All product data is securely stored in the MySQL database and can be retrieved instantly whenever required.

10.2 Objectives of the Product Catalog Module

The objectives of the Product Catalog Module are:

- To maintain complete product information.
 - To simplify inventory management.
 - To reduce manual record keeping.
 - To provide quick product searching.
 - To maintain accurate stock quantities.
 - To support product image management.
 - To improve inventory organization.
 - To ensure secure storage of product data.
-

10.3 Features of the Product Catalog Module

The Product Catalog Module provides the following features:

- Add new products.
- Update existing product information.
- Delete unwanted product records.

Search products by name or category.

Store product images.

Display current stock quantity.

Calculate total stock value.

Validate user input before saving records.

Automatically update the database.

10.4 Product Information Stored

Each product record contains the following information:

Field	Description
Product Id	Unique product identifier
Product Name	Name of the product
Category	Product category
Brand	Manufacturer or brand name
Purchase Price	Cost price of the product
Selling Price	Selling price
Quantity	Available stock quantity
Product Image	Image of the product
Description	Additional product details

This information allows the administrator to manage inventory efficiently and retrieve product details whenever needed.

10.5 Working Process

The Product Catalog Module performs the following operations:

The administrator opens the Product Catalog.

Product details are entered into the form.

The Controller validates all information.

Product data is stored in the MySQL database.

The product list is refreshed automatically.

Products can be searched using the search feature.

Existing records can be updated or deleted whenever required.

Inventory information remains synchronized with the database.

10.6 Product Image Management

The system supports product image management using the Pillow (PIL) library.

Administrators can upload and display product images while creating or editing product records.

Benefits include:

Easy identification of products.

Improved user interface.

Better inventory organization.

Reduced chances of selecting the wrong product.

10.7 Search Functionality

The Product Catalog includes a search feature that allows administrators to locate products quickly.

Products can be searched using:

Product Name

Product ID

Category

Brand

The search function reduces the time required to locate inventory records and improves operational efficiency.

10.8 Product Catalog Screenshot

Figure 10.1: Product Catalog Window

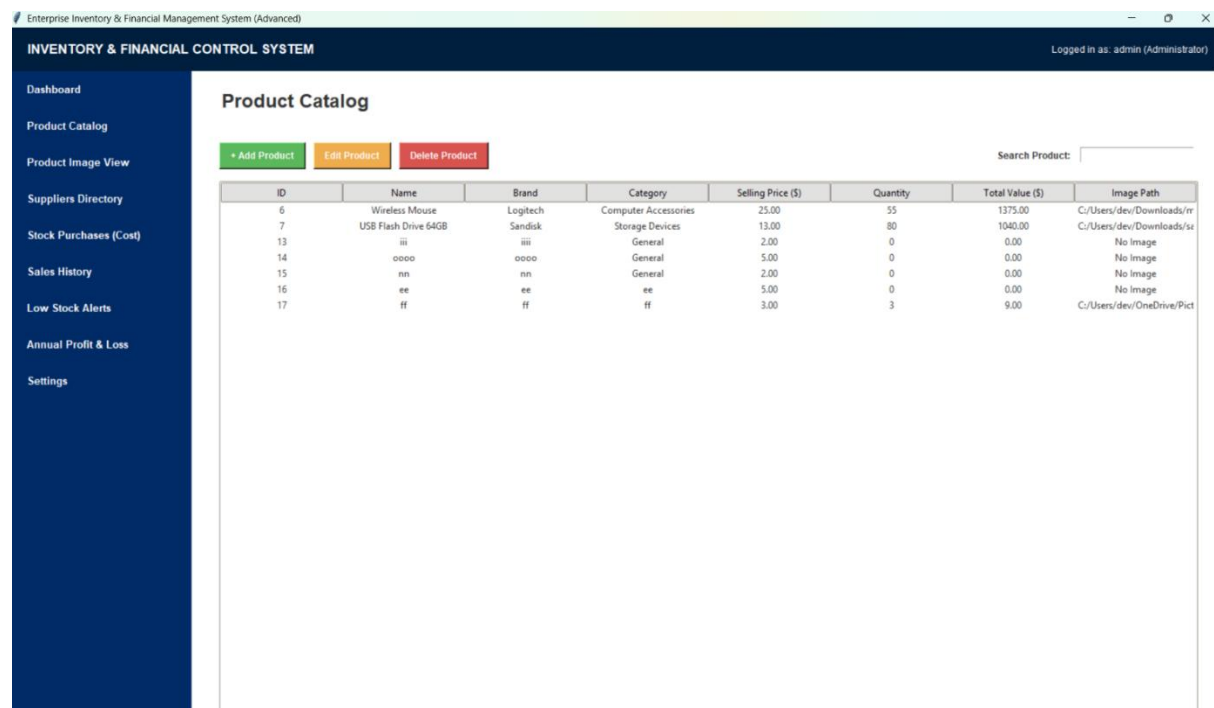


Figure Description

The Product Catalog interface contains:

Product information form.

Product image section.

Search bar.

Product table.

Add button.

Update button.

Delete button.

Clear button.

The interface is designed using Python Tkinter and provides a simple and user-friendly experience for managing inventory.

10.9 Advantages of the Product Catalog Module

The Product Catalog Module provides several advantages:

- Maintains accurate inventory records.
 - Reduces manual paperwork.
 - Supports fast product searching.
 - Improves stock management.
 - Stores product images for easy identification.
 - Minimizes duplicate entries.
 - Enhances inventory organization.
 - Provides secure database storage.
 - Simplifies product updates and maintenance.
-

10.10 Summary

The Product Catalog Module is the foundation of the Enterprise Inventory & Financial Management System. It allows administrators to manage all inventory items efficiently through secure database operations and an easy-to-use graphical interface. Features such as CRUD operations, product image management, and search functionality improve inventory accuracy and enhance overall business productivity.

CHAPTER 11

SUPPLIER DIRECTORY MODULE

11.1 Introduction

The Supplier Directory Module is an essential part of the Enterprise Inventory & Financial Management System. It is designed to maintain complete information about suppliers who provide products to the organization. Proper supplier management ensures smooth procurement of inventory, accurate record keeping, and efficient communication with vendors.

The module allows administrators to add, update, delete, search, and manage supplier information through a simple graphical interface. All supplier details are stored securely in the MySQL database, ensuring quick access and reliable data management.

11.2 Objectives of the Supplier Directory Module

The objectives of this module are:

- To maintain complete supplier information.
 - To simplify supplier management.
 - To reduce manual paperwork.
 - To improve communication with suppliers.
 - To maintain purchase records accurately.
 - To provide quick searching of supplier information.
 - To ensure secure storage of supplier data.
-

11.3 Features of the Supplier Directory Module

The Supplier Directory Module offers the following features:

- Add new supplier records.
- Update existing supplier information.
- Delete supplier records.
- Search suppliers by name or contact number.

Store supplier contact details.

Display supplier purchase history.

Validate user input before saving.

Secure database storage using MySQL.

11.4 Supplier Information Stored

Each supplier record contains the following information:

Field	Description
Supplier Id	Unique supplier identifier
Supplier Name	Name of the supplier or company
Contact Number	Mobile or telephone number
Email Address	Supplier email
Address	Supplier office address
City	Supplier location
State	State or region
GST Number	Tax registration number

These details help the organization maintain proper supplier records and improve procurement management.

11.5 Working Process

The Supplier Directory Module follows these steps:

The administrator opens the Supplier Directory.

Supplier details are entered into the form.

The Controller validates the entered information.

The Model stores the information in the MySQL database.

The supplier list is updated automatically.

Existing supplier records can be edited or deleted.

The search feature retrieves supplier records instantly.

This process ensures accurate and efficient management of supplier information.

11.6 Supplier Directory Screenshot

Figure 11.1: Supplier Directory Window

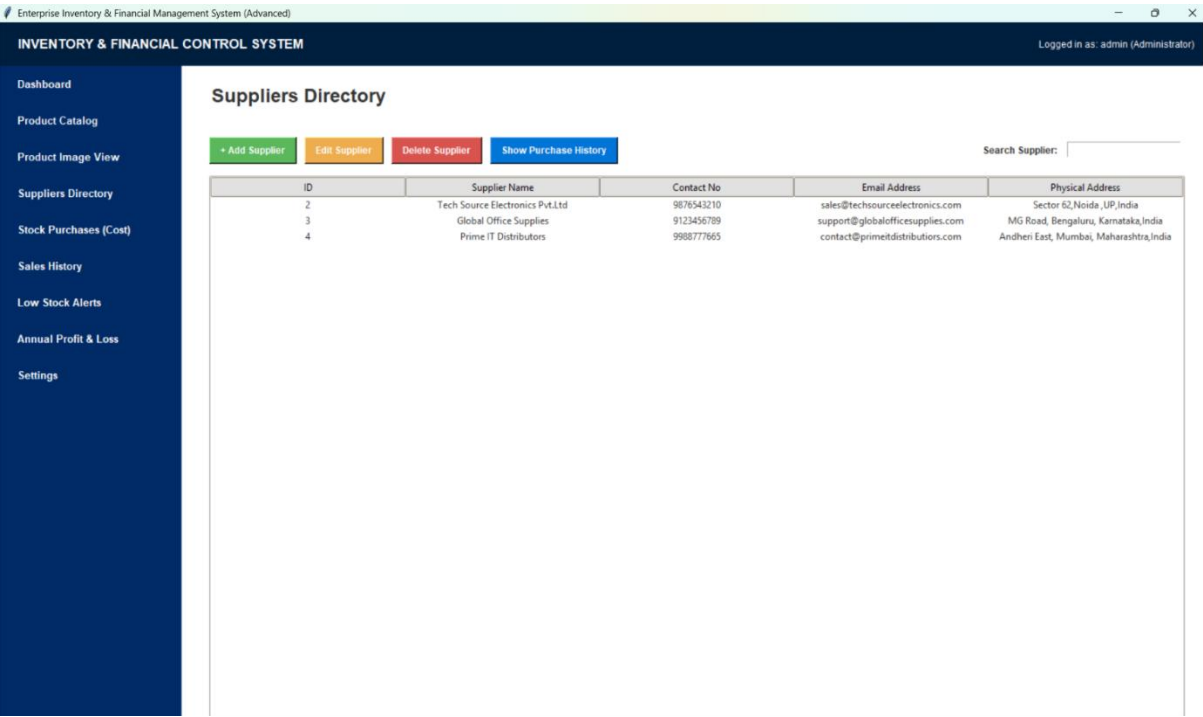


Figure Description

The Supplier Directory interface includes:

Supplier information form.

Search bar.

Supplier table.

Add button.

Update button.

Delete button.

Clear button.

The interface is designed using Python Tkinter and provides an organized and user-friendly environment for managing supplier records.

11.7 Advantages of the Supplier Directory Module

The Supplier Directory Module provides several benefits:

Centralized supplier database.

Quick access to supplier information.

Reduced paperwork.

Improved procurement management.

Faster supplier searching.

Better communication with vendors.

Accurate purchase tracking.

Secure data storage.

Easy maintenance of supplier records.

11.8 Summary

The Supplier Directory Module simplifies the management of supplier information by providing secure storage, fast searching, and easy maintenance of records. It improves procurement efficiency, supports inventory management, and helps organizations maintain strong relationships with suppliers. The module contributes significantly to the overall effectiveness of the Enterprise Inventory & Financial Management System.

CHAPTER 12

STOCK PURCHASE MODULE

12.1 Introduction

The Stock Purchase Module is responsible for managing all product procurement activities in the Enterprise Inventory & Financial Management System. It records every purchase transaction made from suppliers and automatically updates the available stock in the inventory. This module helps organizations maintain sufficient inventory levels and ensures that products are always available to meet customer demand.

By digitizing the purchase process, the system reduces manual calculations, improves inventory accuracy, and maintains a complete history of purchase transactions for future reference.

12.2 Objectives of the Stock Purchase Module

The primary objectives of this module are:

- To record all purchase transactions accurately.
 - To update inventory quantities automatically after every purchase.
 - To maintain detailed purchase records.
 - To calculate purchase expenses automatically.
 - To improve inventory planning.
 - To simplify procurement management.
 - To reduce manual paperwork and calculation errors.
-

12.3 Features of the Stock Purchase Module

The module provides the following features:

- Record new purchase transactions.
- Select products from the inventory database.
- Select suppliers from the supplier directory.
- Enter purchase quantity and cost price.
- Automatically update stock quantity.

Calculate total purchase amount.

Maintain purchase history.

Store purchase date and time.

Validate user input before saving records.

Securely store purchase information in the MySQL database.

12.4 Purchase Information Stored

Each purchase transaction contains the following information:

Field	Description
Purchase ID	Unique transaction identifier
Product Name	Purchased product
Supplier Name	Supplier providing the product
Quantity Purchased	Number of units purchased
Cost Price	Cost per unit
Total Purchase Amount	Total purchase value
Purchase Date	Date of purchase
Purchase Time	Time of purchase
Remarks	Additional notes

This information helps maintain accurate financial records and inventory history.

12.5 Working Process

The Stock Purchase Module follows these steps:

The administrator opens the Stock Purchase Module.

A product is selected from the product list.

The supplier is selected from the supplier directory.

The purchase quantity and cost price are entered.

The Controller validates all entered information.

The purchase record is saved in the MySQL database.

The available stock quantity is updated automatically.

Purchase expenses are included in financial calculations.

The Dashboard is refreshed to display updated statistics.

12.6 Automatic Stock Update

One of the major features of this module is the automatic stock update mechanism.

Whenever a purchase transaction is completed:

The purchased quantity is added to the existing stock.

Product availability is updated immediately.

Inventory records remain synchronized.

Dashboard statistics are refreshed.

Future sales use the updated stock quantity.

This automation minimizes manual intervention and ensures inventory accuracy.

12.7 Purchase Module Screenshot

Figure 12.1: Stock Purchase Window

Figure Description

The Stock Purchase interface contains:

Product selection.

Supplier selection.

Purchase quantity field.

Cost price field.

Total amount calculation.

Save button.

Update button.

Purchase history table.

The interface is developed using Python Tkinter and provides a simple and efficient environment for managing inventory purchases.

12.8 Advantages of the Stock Purchase Module

The Stock Purchase Module provides several benefits:

Maintains complete purchase history.

Updates inventory automatically.

Improves stock accuracy.

Reduces manual calculations.

Supports financial reporting.

Simplifies supplier management.

Prevents inventory shortages.

Improves procurement efficiency.

Provides secure database storage.

12.9 Summary

The Stock Purchase Module plays an important role in maintaining inventory availability and accurate financial records. By recording every purchase transaction, updating stock automatically, and storing complete purchase history, the module improves inventory control and supports effective business management. Its integration with the Dashboard and financial modules ensures that all business information remains accurate, synchronized, and readily available for decision-making.

CHAPTER 13

SALES HISTORY MODULE

13.1 Introduction

The Sales History Module is an important component of the Enterprise Inventory & Financial Management System. It is responsible for recording and managing all product sales performed by the organization. Every sales transaction is stored securely in the MySQL database, allowing administrators to monitor sales performance, calculate revenue, and maintain complete transaction records.

The module automatically reduces the available stock whenever a sale is completed, ensuring that inventory quantities remain accurate. It also contributes to financial analysis by providing data required for revenue calculation and annual Profit & Loss reports.

13.2 Objectives of the Sales History Module

The objectives of this module are:

- To record every sales transaction accurately.
 - To maintain complete sales history.
 - To reduce inventory automatically after each sale.
 - To calculate revenue automatically.
 - To improve inventory accuracy.
 - To support financial reporting.
 - To simplify sales management.
-

13.3 Features of the Sales History Module

The module provides the following features:

- Record product sales.
- Search sales records.
- Display customer purchase information (if applicable).
- Automatically reduce stock quantity.

Calculate total sales amount.

Maintain complete sales history.

Store transaction date and time.

Generate sales reports.

Secure database storage using MySQL.

13.4 Sales Information Stored

Each sales transaction contains the following information:

Field	Description
Sale ID	Unique sales transaction number
Product Name	Name of the sold product
Quantity Sold	Number of units sold
Selling Price	Price per unit
Total Amount	Total sale value
Sale Date	Date of sale
Sale Time	Time of sale
Customer Name	Customer details
Payment Method	Cash, UPI, Card, etc.

13.5 Working Process

The Sales History Module performs the following operations:

The administrator opens the Sales Module.

A product is selected from the inventory.

The quantity to be sold is entered.

The Controller validates stock availability.

The sale is recorded in the MySQL database.

Available stock is reduced automatically.

Revenue is updated instantly.

Dashboard statistics refresh automatically.

Sales history is available for future reference.

13.6 Sales History Screenshot

Figure 13.1: Sales History Window

Figure Description

The Sales History interface contains:

Product selection.

Sales quantity field.

Selling price.

Total amount calculation.

Save button.

Search option.

Sales history table.

The interface provides a simple and efficient way to manage product sales while maintaining accurate inventory records.

13.7 Advantages of the Sales History Module

The Sales History Module provides several benefits:

Maintains complete sales records.

Automatically updates inventory.

Calculates revenue accurately.

Reduces manual calculations.

Improves financial reporting.

Supports business analysis.

Provides secure storage of sales transactions.

Simplifies inventory tracking.

13.8 Summary

The Sales History Module records every sales transaction and automatically updates inventory levels after each sale. It ensures accurate revenue calculation, maintains complete sales records, and supports financial analysis. By integrating with the Dashboard and Profit & Loss Module, it contributes to effective inventory management and informed business decision-making.

CHAPTER 14

LOW STOCK ALERT MODULE

14.1 Introduction

The Low Stock Alert Module is an intelligent feature of the Enterprise Inventory & Financial Management System. It continuously monitors the quantity of products available in inventory and automatically identifies items whose stock falls below the predefined minimum level. This feature helps administrators take timely action to replenish inventory and prevents stock shortages.

By providing automatic alerts, the system minimizes the risk of running out of essential products and supports uninterrupted business operations.

14.2 Objectives of the Low Stock Alert Module

The objectives of this module are:

- To monitor inventory levels continuously.
 - To identify products with low stock automatically.
 - To generate alerts for products requiring replenishment.
 - To support timely purchasing decisions.
 - To prevent stock shortages.
 - To improve inventory planning and management.
 - To reduce the possibility of lost sales due to unavailable products.
-

14.3 Features of the Low Stock Alert Module

The module provides the following features:

- Automatic monitoring of stock quantities.
- Identification of products below the minimum stock level.
- Dashboard notification for low-stock items.
- Purchase request generation.
- Real-time inventory updates.
- Integration with the Product Catalog and Dashboard modules.

Accurate and reliable stock monitoring.

14.4 Working Process

The Low Stock Alert Module performs the following steps:

The system monitors product quantities stored in the MySQL database.

Each product quantity is compared with the predefined minimum stock level.

If the available quantity falls below the minimum level, the product is marked as **Low Stock**.

A notification is displayed on the Dashboard.

The administrator reviews the alert and creates a purchase order if required.

After new stock is purchased, the inventory quantity is updated automatically, and the alert is removed.

This automated process ensures that inventory is replenished before products become unavailable.

14.5 Dashboard Notification

Whenever a product reaches the minimum stock level, the Dashboard displays a notification containing:

Product Name

Available Quantity

Minimum Required Quantity

Stock Status

Purchase Request Indicator

These notifications help administrators identify products that require immediate attention.

14.6 Low Stock Alert Screenshot

Figure 14.1: Low Stock Alert Window

Figure Description

The Low Stock Alert interface displays:

List of products with low stock.

Current available quantity.

Minimum stock level.

Alert status.

Purchase request option.

The interface provides administrators with real-time information about inventory shortages, enabling timely procurement decisions.

14.7 Advantages of the Low Stock Alert Module

The Low Stock Alert Module provides several advantages:

Prevents inventory shortages.

Improves inventory planning.

Reduces the risk of lost sales.

Supports timely purchasing decisions.

Provides real-time inventory monitoring.

Enhances operational efficiency.

Reduces manual stock checking.

Improves customer satisfaction by ensuring product availability.

14.8 Summary

The Low Stock Alert Module plays a vital role in maintaining optimal inventory levels. By automatically monitoring stock quantities and generating timely notifications, it helps administrators replenish products before they run out. This feature improves inventory control, reduces operational risks, and contributes to efficient business management.

CHAPTER 15

ANNUAL PROFIT & LOSS MODULE

15.1 Introduction

The Annual Profit & Loss Module is one of the most important financial components of the Enterprise Inventory & Financial Management System. It provides administrators with a comprehensive overview of the organization's financial performance by calculating annual revenue, total expenses, and net profit. The module automatically gathers financial information from purchase and sales records stored in the MySQL database and generates an accurate financial summary.

This module eliminates manual financial calculations, reduces errors, and enables business owners to evaluate the overall performance of their organization. The generated financial information supports better planning, budgeting, and strategic decision-making.

15.2 Objectives of the Annual Profit & Loss Module

The objectives of this module are:

To calculate annual business revenue automatically.

To calculate total annual expenses.

To determine annual net profit.

To generate financial reports for analysis.

To improve financial decision-making.

To eliminate manual financial calculations.

To provide accurate and reliable financial information.

15.3 Features of the Annual Profit & Loss Module

The module provides the following features:

Automatic revenue calculation.

Automatic expense calculation.

Net profit calculation.

Annual financial summary.

Dashboard integration.

Real-time financial updates.

Secure storage of financial records.

Easy-to-read financial reports.

15.4 Revenue Calculation

Revenue represents the total income earned from product sales during the financial year.

The system automatically calculates annual revenue by adding the value of all sales transactions recorded in the Sales History Module.

Formula:

Total Revenue = Sum of All Sales Transactions

The calculated revenue is displayed on the Dashboard and included in the annual financial report.

15.5 Expense Calculation

Expenses represent the total amount spent on purchasing products and other inventory-related costs.

The system automatically calculates total expenses using purchase transaction records stored in the database.

Formula:

Total Expenses = Sum of All Purchase Transactions

This calculation provides administrators with an accurate record of annual business expenditure.

15.6 Net Profit Calculation

Net Profit is calculated by subtracting total expenses from total revenue.

Formula:

Net Profit = Total Revenue – Total Expenses

The calculated net profit helps administrators determine whether the business has earned a profit or incurred a loss during the financial year.

15.7 Working Process

The Annual Profit & Loss Module performs the following operations:

The system retrieves sales records from the Sales History Module.

The system retrieves purchase records from the Stock Purchase Module.

Total annual revenue is calculated.

Total annual expenses are calculated.

Net profit is computed automatically.

Financial information is displayed on the Dashboard.

Annual financial reports are generated for business analysis.

This automated workflow ensures accurate and timely financial reporting.

15.8 Annual Profit & Loss Screenshot

Figure 15.1: Annual Profit & Loss Dashboard

Dashboard	Annual Profit & Loss					
Product Catalog	Annual Profit & Loss Statement					
Product Image View	Total Revenue (Sales): \$2426.00					
Suppliers Directory	Total Expenses (Purchases): \$1306.00					
Stock Purchases (Cost)	Net Profit / Loss: \$1120.00					
Sales History	Revenue Breakdown (Sales Transactions)					
Low Stock Alerts	ID	Item Name	Selling Price (\$)	Quantity Sold	Total Revenue (\$)	Date & Time
Annual Profit & Loss	10	wireless keyboard	100.00	10	1000.00	2026-07-27 13:33:46
Settings	9	wireless keyboard	100.00	10	1000.00	2026-07-27 13:33:30
	8	milk	10.00	12	120.00	2026-07-27 13:31:43
	7	milk	5.00	12	60.00	2026-07-27 13:25:29
	6	oooo	6.00	6	36.00	2026-07-27 13:22:56
	5	iii	5.00	10	50.00	2026-07-27 13:19:19
	4	oooo	5.00	6	30.00	2026-07-27 13:14:59
	3	nn	5.00	14	70.00	2026-07-27 13:14:23
	2	ee	10.00	2	20.00	2026-07-27 13:10:17
	1	almond oil	2.00	20	40.00	2026-07-22 02:14:30
	Expense Breakdown (Stock Purchases)					
	ID	Item Name	Supplier Name	Cost Price (\$)	Quantity	Date & Time
	1	almond oil	sfa	2.00	20	2026-07-22 02:12:56
	2	almond oil	egdsf	2.00	20	2026-07-22 02:13:49
	3	as	fdcgvb	2.00	20	2026-07-22 10:18:46
	4	ds	uhh	3.00	30	2026-07-22 21:32:00
	5	uhb	jijbjnm	20.00	10	2026-07-22 21:32:34
	6	wireless mouse	global tech supplies	25.00	20	2026-07-25 21:29:00
	7	ds	sd	3.00	4	2026-07-25 21:46:29
	8	milk	nhu	5.00	2	2026-07-27 12:27:30
	9	milk	njk	2.00	10	2026-07-27 12:28:04
	10	milk	km	2.00	2	2026-07-27 12:28:31
	11	milk	nik	2.00	10	2026-07-27 12:28:53

Figure Description

The Profit & Loss interface displays:

Annual Revenue

Annual Expenses

Net Profit

Financial Summary

Revenue and Expense Comparison

Business Performance Indicators

This information enables administrators to evaluate the financial health of the organization quickly and accurately.

15.9 Advantages of the Annual Profit & Loss Module

The Annual Profit & Loss Module provides several advantages:

Eliminates manual financial calculations.

Generates accurate financial reports.

Improves business planning.

Supports strategic decision-making.

Provides real-time financial analysis.

Reduces accounting errors.

Saves time and effort.

Enhances financial transparency.

15.10 Summary

The Annual Profit & Loss Module provides an efficient method for evaluating the financial performance of the organization. By automatically calculating annual revenue, expenses, and net profit, the system generates reliable financial reports that support effective business management. This module enhances financial accuracy, reduces manual workload, and enables administrators to make informed decisions based on real-time business data.

CHAPTER 16

SETTINGS MODULE

16.1 Introduction

The Settings Module is an important utility component of the Enterprise Inventory & Financial Management System. It enables administrators to customize system preferences, manage account settings, and improve the overall usability of the application. Through this module, users can change themes, update passwords, configure application preferences, and securely log out of the system.

The module enhances both security and user experience by providing personalization options while ensuring that only authorized users can modify sensitive settings.

16.2 Objectives of the Settings Module

The objectives of the Settings Module are:

To provide system customization options.

To allow secure password management.

To improve application usability.

To support theme customization.

To protect user accounts.

To provide secure logout functionality.

To improve overall user experience.

16.3 Features of the Settings Module

The Settings Module provides the following features:

Change application theme.

Update login password.

Secure logout.

User account management.

System preference configuration.

Easy navigation back to the Dashboard.

User-friendly interface.

16.4 Theme Management

The application provides theme customization to improve user comfort and accessibility.

Light Theme

The Light Theme offers a bright interface suitable for well-lit environments. It provides clear visibility, improves readability, and gives the application a professional appearance.

Dark Theme

The Dark Theme uses darker background colors to reduce eye strain during extended periods of use. It is particularly useful in low-light environments and provides a modern look and feel.

Administrators can switch between themes according to their preferences.

16.5 Password Management

The Settings Module allows administrators to update their login password whenever required.

The password update process includes:

Enter the current password.

Enter the new password.

Confirm the new password.

Validate the information.

Save the updated password in the MySQL database.

This process improves account security and helps protect sensitive business information.

16.6 Logout Feature

The Logout feature enables users to exit the application securely.

When the administrator selects the Logout option:

The current session ends.

User data is protected.

The Login screen is displayed again.

Unauthorized access is prevented.

Secure logout is an essential feature for maintaining application security.

16.7 Working Process

The Settings Module operates as follows:

The administrator opens the Settings Module.

The desired setting is selected.

Changes are made to the selected option.

The Controller validates the entered information.

Updated settings are saved successfully.

Confirmation is displayed to the administrator.

This process ensures that all changes are applied safely and correctly.

16.8 Settings Module Screenshot

Figure 16.1: Settings Window

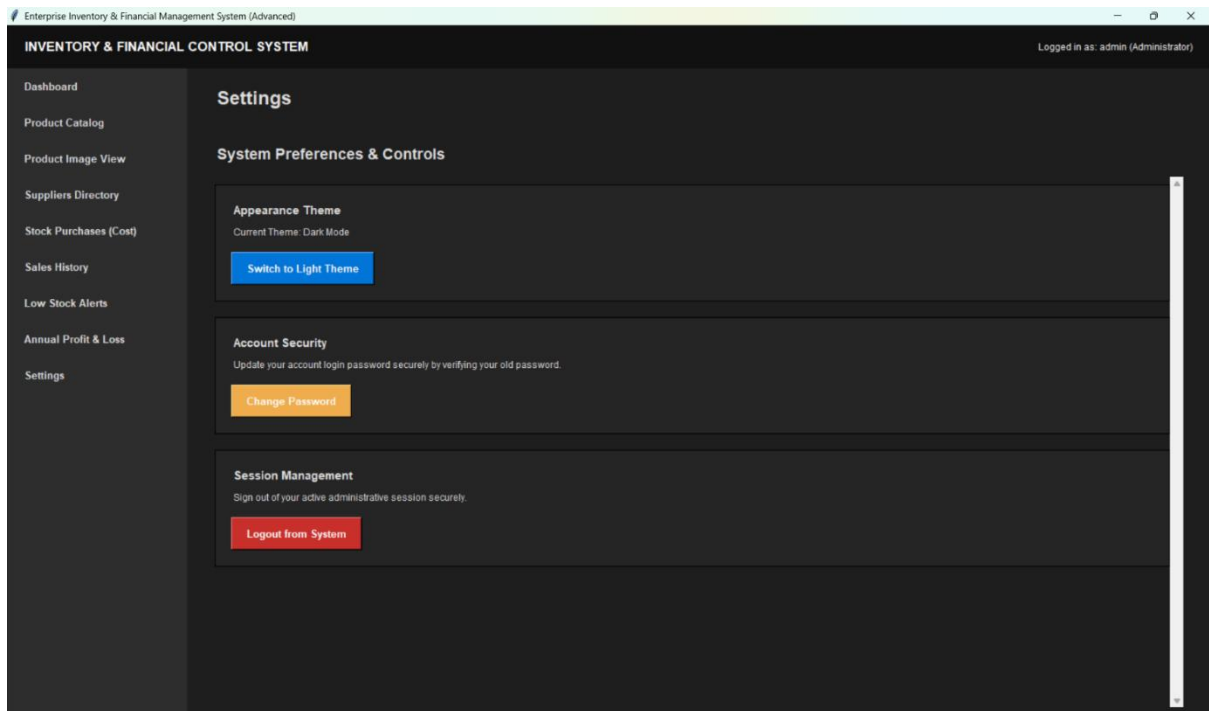


Figure Description

The Settings interface contains:

Theme selection options.

Change Password section.

Save button.

Cancel button.

Logout button.

The interface is designed to be simple, secure, and easy to navigate.

16.9 Advantages of the Settings Module

The Settings Module provides several advantages:

Improves account security.

Allows password updates.

Supports personalized themes.

Enhances user experience.

Provides secure logout.

Simplifies application customization.

Easy to use and maintain.

Improves overall system usability.

16.10 Summary

The Settings Module provides administrators with essential customization and security features. It allows users to manage their accounts, personalize the application interface, and securely log out of the system. These capabilities improve the overall usability, flexibility, and security of the Enterprise Inventory & Financial Management System while ensuring a better user experience.

CHAPTER 17

ADVANTAGES OF THE SYSTEM

17.1 Introduction

The Enterprise Inventory & Financial Management System provides a modern and efficient solution for managing inventory, suppliers, purchases, sales, and financial records. By replacing manual record keeping with a computerized system, the application improves operational efficiency, reduces errors, and supports better business decision-making. The system integrates multiple business functions into a single platform, making inventory management easier and more reliable.

17.2 Advantages

Improved Inventory Management

The system maintains accurate records of all products and automatically updates stock quantities after every purchase and sale transaction. This minimizes manual errors and ensures that inventory information remains current.

Centralized Data Management

All business information, including product details, supplier records, purchase transactions, sales history, and financial reports, is stored securely in a centralized MySQL database. This enables quick access to information and simplifies data management.

Increased Accuracy

Automated calculations reduce human errors associated with manual inventory and financial management. Revenue, expenses, and net profit are calculated automatically, ensuring reliable business reports.

Time Saving

Routine operations such as adding products, searching records, updating inventory, and generating reports are completed quickly. This reduces administrative workload and improves productivity.

Enhanced Security

The Login Module ensures that only authorized users can access the application. Password authentication and controlled database access protect sensitive business information.

Low Stock Monitoring

The system continuously monitors inventory levels and generates alerts whenever product quantities fall below the minimum stock level. This helps prevent stock shortages and supports timely purchasing decisions.

Financial Analysis

The Annual Profit & Loss Module automatically calculates revenue, expenses, and net profit, providing valuable insights into business performance and supporting financial planning.

User-Friendly Interface

The application is developed using Python Tkinter, providing a simple, attractive, and easy-to-use graphical interface that requires minimal technical knowledge.

Modular Design

The MVC Architecture organizes the application into separate components, making the system easier to maintain, debug, and enhance in the future.

Scalability

The system can be extended by adding new modules such as barcode scanning, GST billing, cloud synchronization, mobile applications, and business analytics without major modifications.

17.3 Summary

The Enterprise Inventory & Financial Management System offers an efficient, secure, and scalable solution for inventory and financial management. Its automation features reduce manual work, improve accuracy, and provide reliable business information, making it suitable for educational purposes as well as small and medium-sized organizations.

CHAPTER 18

CONCLUSION

18.1 Conclusion

The Enterprise Inventory & Financial Management System (Advanced) has been successfully designed and developed using Python, Tkinter, MySQL, and the Model–View–Controller (MVC) architecture. The project automates key business operations, including inventory management, supplier management, purchase tracking, sales recording, financial reporting, and low-stock monitoring.

The application demonstrates how modern software technologies can simplify inventory management, improve data accuracy, and reduce manual effort. Automatic calculations, centralized database storage, secure user authentication, and an intuitive graphical interface contribute to efficient business operations and better decision-making.

This project also provided practical experience in desktop application development, database management, object-oriented programming, and software engineering principles. It successfully meets the objectives defined at the beginning of the project and serves as a reliable solution for inventory and financial management.

18.2 Future Scope

The project can be further enhanced by implementing additional features such as:

Barcode and QR code scanning.

GST invoice generation.

Customer management.

Mobile application support.

Cloud database integration.

Multi-user role management.

Email and SMS notifications.

Advanced sales analytics and dashboards.

AI-based inventory demand forecasting.

Automatic data backup and recovery.

These enhancements would make the system more powerful, scalable, and suitable for large-scale business environments.

REFERENCES

Python Software Foundation. *Python Documentation*. <https://docs.python.org/>

Oracle Corporation. *MySQL Documentation*. <https://dev.mysql.com/doc/>

Python Tkinter Documentation. <https://docs.python.org/3/library/tkinter.html>

Pillow Documentation. <https://pillow.readthedocs.io/>

Microsoft Visual Studio Code Documentation. <https://code.visualstudio.com/docs>

BIBLIOGRAPHY

Herbert Schildt, *Python Programming*.

Reema Thareja, *Python Programming Using Problem Solving Approach*.

Ian Sommerville, *Software Engineering*.

Elmasri & Navathe, *Fundamentals of Database Systems*.

Silberschatz, Korth & Sudarshan, *Database System Concepts*.

APPENDIX

Appendix A – Software Screenshots

Login Module

Dashboard Module

Product Catalog Module

Supplier Directory Module

Stock Purchase Module

Sales History Module

Low Stock Alert Module

Annual Profit & Loss Module

Settings Module

Appendix B – Source Code

Include the important Python source code files, MVC folder structure, database schema, and SQL scripts used for developing the Enterprise Inventory & Financial Management System.

Appendix C – Database Tables

Users

Products

Suppliers

Purchases

Sales

Include table structures and sample records if required by your institution.

